

The Preprint Set — All Eight on One Sheet

*P-branch publication set · docs/research/*_preprint.{tex,pdf} · one house layout · author Yigit Onat <yionat@gmail.com>*

P1 polish → P2 SSRN → P3 arXiv q-fin.TR → P4 outreach → P5 track · EPFL EDFI deadlines **Jan 15 / Mar 31 2027** · card built 2026-08-08

1 convolver — Event-Aligned SVD 15pp

Analytically-defined events (breakout / turtle-soup / trap) × 4 channels (body, wicks, volume) → SVD → IC gate with BH-FDR. **SVD never sees forward returns** — the structural anti-overfit argument. 7.2M ticks, 12,059 candles, 1,775 events. **6 of 158 kernels survive FDR**, $|IC| \in [0.20, 0.36]$; **no breakout kernel survives**. Rolling stability $\bar{\rho} > 0.90$ for the strongest. [!] No kernel passes OOS robustness — two sign-flip. Method is the result. *P1 camera-ready target. Lead for Teichmann, Bölskei, Malamud, Cheridito, Zdeborová, Krzakala, Filipović.*

2 microstructure_alpha — FDR-Controlled Discovery 12pp

236 features, 100 ms, BTC/ETH/SOL, 1 s–15 min. Imbalance dominates, $IC \approx 0.45$ at 1–5 s, universal across symbols, 7/8 robustness (BTC). **Central negative result**: conditioning on directionally-correct passive fills collapses $IC\ 0.45 \rightarrow 0.03$ on all three symbols — a structural **adverse-selection barrier**; execution, not signal quality, binds. Five-gate protocol (discovery/cost/temporal/cross-symbol/dedup) + BH-FDR $q=0.05$ + adaptive threshold + decay monitoring. [!] Combiner $IC\ .18/.25/.36$ on a thin window — those algorithms were REFUTED by the Q4 kill gate 2026-07-30. The barrier result is unaffected. *Lead for Collin-Dufresne, Hugonnier, Acciaio, Bühlmann, Farkas, Kuhn.*

3 liquidity_heatmap — Liquidation Geometry 11pp

$K=200$ bins, 10 bps, $\pm 10\%$; 8 scale-invariant features. Domino condition $\delta_B - \delta_A < \lambda M_A/m_t$, depth as damper, sign(ACP) predicts sign. **No performance numbers anywhere — deliberate**. Aug-2026 revision (§7): sampling rate now *measured* (coverage .315/.380/.247 at $s=2$), and a **second bias found** — 2.9% of liquidated wallets vs 21.9% of liquidated notional; the OI rescaling cannot repair it (assumes uniform thinning; this is biased in shape). [!] Mapped frac 0.0% is confounded by snapshot ordering, not a refutation.

4 prism — Perception Pressure & Resonance 7pp

$P(K, t) = A(K, t) \cdot V(K, t)$: salience ≥ 0 times credibility-weighted valence in $[-1, +1]$. Three properties novel in combination — multiplicative coupling (signed, magnitude-meaningful), **diversity as multiplier** (one outlet cannot inflate it), full decomposability. Trust *earned* by automated cross-corroboration, $0.1 + 0.4 \cdot (\text{validated}/\text{total})$, never hand-assigned. Five-agent pipeline. System at $\sim$ Prism. [!] Methods/system paper; no empirics logged. *Lead for Leippold — send with #5.*

5 prism_signal — From Narrative to Alpha 9pp

What does one exogenous signal add to a structurally narrow microstructure book? Three reasons it helps without being individually strong: **timescale separation** (disjoint from the 0.005–0.1 Hz imbalance band), **effective breadth** ($N_{\text{eff}} = N/[1 + (N-1)\bar{\rho}]$ — a weak orthogonal signal beats a strong redundant one), **partial evasion** of the adverse-selection barrier (patient execution couples weakly to the fillable-state indicator). Same as-of join, same five gates, no privileges. [!] All three stated as hypotheses with FDR-corrected tests; no performance figures. Weak point: $\bar{\rho}$ must be estimated under *stress*, per #6.

6 process — The Process as an Analytical Unit 14pp

Third first-class unit beside feature and algorithm: a typed, registry-bound contract certifying whether something carries information about a precisely specified target. MI as currency (justified by **Fano**, a hard lower bound on error). $IR \approx \overline{IC}\sqrt{N_{\text{eff}}}$; **the binding correlation is the stress-regime one, not the calm one**. Five mechanisms: CMI greedy selection, interaction-information synergy mining, prefix residualization, Marchenko–Pastur-denoised PCs, capstone decorrelated book reporting honest N_{eff} . Aspect×horizon target taxonomy; derivative operators under a program-level FDR ledger. [!] First real surface: 284 BTC cells, 0 discoveries — BH at the 100-shuffle p -floor *cannot* pass. Power problem, not a verdict.

7 lob_predictive_structure — Two Channels 8pp

2.17M ticks/symbol. **Two empirically orthogonal channels**: directional (IC to 0.47 at 1–5 s, 8 independent axes, imbalance-led) and volatility (IC 0.35, arrival intensity/activity/ toxicity) — each carries *zero* of the other’s information, on all three symbols. Spectral localization to 0.005–0.1 Hz, OU half-life 5–7 s, cross-coherence 68 s; **time-**

domain aggregation destroys the signal. Efficiency drift measured live (5 s $IC\ .45 \rightarrow .29$ as spreads tighten). Execution boundary: maker-fill conditioning $\rightarrow 0.03$, but **any-fill conditioning** $\rightarrow 0.52$ — which isolates adverse selection rather than decay. [!] OOS replication in progress, not done.

8 nat2_exact_map — Exact or Nothing 11pp

Instrument for #3. Two rules: **exact or nothing** (unobserved $\Rightarrow$ absent + coverage deficit, never imputed) and **gates before models** (feed $\rightarrow$ map $\rightarrow$ magnet $\rightarrow$ persistence $\rightarrow$ decay; the refusal is the product). WORM store, dual clocks $t^{\text{event}}/t^{\text{ingest}}$ so lookahead is *queryable*; hash-chained ledger (threat model is the researcher, not an attacker); admissibility = exists $\wedge$ passed $\wedge$ fresh. **gate feed** PASS 20/20; **gate map FAIL** on **predictive** — an unevaluable criterion must not pass. Coverage .315/.380/.247; derivation exact for 40.8% of 2,926. Code at $\sim$ nat2 (separate repo).

Cross-map — who supplies what

- **The barrier**: #2 finds it, #7 isolates it (0.03 vs 0.52), #5 proposes to evade it, #8 builds around it.
- **Breadth** / N_{eff} : #6 is the theory, #7’s 8 axes are an instance, #5 is the argument for spending it on a new domain.
- **Liquidation cascades**: #3 is the model, #8 is the instrument and revises #3’s obstacle.
- **Narrative**: #4 is the instrument, #5 is the integration theory.
- **Standalone**: #1 (method paper; different mathematics, best fit for the D-MATH/D-ITET supervisors).

Posture shared by 3, 4, 5, 6, 8

Specify the instrument, expose the constraint, **report no fabricated performance**, state the falsifying experiment before running it. Five of eight papers report a negative or unevaluable result as their headline. That is the set’s signature and should be foregrounded in outreach, not apologised for.

Outreach map — lead preprint per supervisor

Tier 1 (send first).

Teichmann (<i>ETH D-MATH</i>)	#1 convolver
Collin-Dufresne (<i>EPFL SFI</i>)	#2 micro_alpha
Hugonnier (<i>EPFL SFI</i>)	#2 micro_alpha
Malamud (<i>EPFL SFI</i>)	#1 convolver
Bölskei (<i>ETH D-ITET</i>)	#1 convolver

Tier 2 (stagger +2 weeks).

Acciaio (<i>ETH D-MATH</i>)	#2
Cheridito (<i>ETH D-MATH</i>)	#1
Bühlmann (<i>ETH D-MATH</i>)	#2
Farkas (<i>ETH/UZH</i>)	#2
Leippold (<i>UZH/SFI</i>)	#5 (with #4)
Zdeborová (<i>EPFL SPOC</i>)	#1
Krzakala (<i>EPFL IdePHICS</i>)	#1
Kuhn (<i>EPFL RAO</i>)	#2
Filipović (<i>EPFL Swissquote</i>)	#1

[!] #7 is arguably the better lead than #2 for the pure-microstructure/econometrics targets (Collin-Dufresne, Hugonnier, Bühlmann) — it is measurement-first rather than apparatus-first, and it does not carry the refuted combiner numbers. Not yet reassigned in docs/PHD/README.md.

Co-supervision pairings

Teichmann + Bölskei (ML in finance + signal-processing theory) · Collin-Dufresne + Krzakala (microstructure + random matrix theory) · Malamud + Zdeborová (ML asset pricing + statistical physics).

Before sending — known debts

- #2 §Hierarchical combiner rests on the five algorithms REFUTED 2026-07-30. Re-check or caveat before P1 camera-ready.
- #7’s out-of-time replication is stated as in progress; make sure the version sent says so.
- docs/TASKS.md P1 still reads “camera-ready *convolver* PDF” — the set is now eight.
- Prof email addresses are *not* stored anywhere; gathering them is part of P4.